

CRedit Author Contribution Statement

OligoVigil submission to Nucleic Acids Research, Database Issue

2026-06-11

CRedit Author Contribution Statement

This statement follows the Contributor Roles Taxonomy (CRedit) as adopted by Nucleic Acids Research and Oxford University Press. The 14 standard CRedit roles are: Conceptualisation, Data Curation, Formal Analysis, Funding Acquisition, Investigation, Methodology, Project Administration, Resources, Software, Supervision, Validation, Visualisation, Writing — Original Draft, and Writing — Review & Editing.

Role assignments below follow the CRedit taxonomy and have been prepared for author confirmation before submission.

Jie Ni — Conceptualisation; Methodology; Software; Investigation; Formal analysis; Data curation; Visualisation; Writing — original draft; Project administration.

Xinting Zhang — Investigation; Data curation; Validation.

Zhuoying Xie — Investigation; Visualisation.

Shan Lu — Resources; Writing — review & editing.

Yun Liu — Conceptualisation; Supervision; Writing — review & editing; Resources.

Adam Jatowt — Supervision; Writing — review & editing; Resources.

Notes specific to OligoVigil

- **Data Curation (Jie Ni).** Sole human curator of record (`curator_id = 'ni_jie'` in `curation_audit`). Adjudicated the 2,003-candidate proposal pool and signed off on curator-verified release decisions and grade assignments.
- **Validation (Jie Ni, with Xinting Zhang).** Designed and executed the machine-stage false-accept audit (66/90 false accepts among v1 accept calls; Wilson 95% CI [0.63, 0.81]), the candidate-demotion audit, the release-row audit reconciliation and the second-curator reliability analyses reported in the manuscript.
- **Methodology (Jie Ni).** Designed the three-stage candidate-then-model-then-human architecture with the candidate-vs-release firewall; defined the toxicity and off-target benchmark splits; and specified the pair-level source-by-molecule isolation invariant.
- **Software (Jie Ni).** Implemented the curation pipeline, the 70,283-task work queue, the v2 LLM-curator under exclusion-first + verbatim-grounding-quote constraints, the QA suites (`smoke_test.py`, `frontend_contract_check.py`, `final_delivery_check.py`), the public web portal (no-login REST API + OpenAPI + MCP + Bioschemas + W3C PROV), and the release scripts that produced

evidence_release.csv, benchmark_reference_splits.csv, curation_audit.csv, and v2_human_override_decisions.csv.

- **Writing - Original Draft (Jie Ni).** Drafted the manuscript and assembled the numerical claim audit supporting the reported release, benchmark and validation counts.
- **Writing - Review & Editing (all).** Reviewed and revised the manuscript. All authors are responsible for the final wording.
- **Supervision (Yun Liu, Adam Jatowt).** Provided scientific oversight and reviewed major design decisions including the candidate/release firewall and curation-integrity framework.
- **Resources (Shan Lu, Yun Liu, Adam Jatowt).** Provided institutional, domain and research-computing resources used to run the curation workflows and QA suites; the computational work used the Austrian Scientific Computing (ASC) federated MUSICA cluster and the LEO5 high-performance computing facility at the University of Innsbruck (see 04_funding.md for the full acknowledgement).

Author identities (locked, ORCID-validated)

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All six ORCID iDs have been checksum-validated (ISO 7064 MOD 11-2).